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Network analysis of kick-in possession chains in elite Australian football

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ABSTRACT

The study aim was to investigate ball movement patterns using network analysis techniques, to compare between successful and unsuccessful outcomes and teams in the Australian Football League (AFL). This analysis focused on possession chains starting from a kick-in ($n = 1,720$), drawn from all games played in the 2015 AFL Premiership season (18 teams, 206 games). Player interactions were quantified using four network metrics: cluster coefficient, degree centrality, network density, and entropy. Three-way ANOVA with Tukey post hoc and ω^2 effect sizes were calculated to assess whether differences existed between kick-in outcomes, ladder brackets, and match outcomes for each network metric. No significant differences were observed between ladder brackets or match outcomes for any network metric. More successful kick-in chains were characterised by lower density ($\omega^2 = 0.26$, large effect; $F(9, 1678) = 66.6$, $p < 0.00$) and higher entropy ($\omega^2 = 0.17$, large effect; $F(9, 1678) = 39.6$, $p < 0.00$). This suggests that chains resulting in successful kick-in outcomes exhibited lower interconnectedness, with a high number of players involved, and lower predictability in ball movement patterns. These findings have practical value for coaches and performance analysts and support further applications of network analysis in Australian football.

ARTICLE HISTORY

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KEYWORDS

Network analysis;
performance analysis;
Australian football; elite
sport

Introduction

In much of the existing literature, team success has been assessed through simplistic match outcomes, for example, matches won or lost (Robertson et al., 2016), and by analysing a range of aggregated individual and team key performance indicators (KPIs) (Ortega et al., 2009; Parejo et al., 2013; Sgro et al., 2015). While individual player KPIs can provide some insight into performance, collective team performance cannot be inferred by simply combining the actions measured through individual performance metrics, as interactions between individuals are influential in determining a team's success or otherwise (Bourbousson et al., 2010; Fewell et al., 2012; Trequattrini et al., 2015). Consequently, recent research in performance analysis has focused on understanding the collective actions and player interactions that are associated with and lead to successful team outcomes (Clemente, Lourenço Martins et al., 2015; Peña & Touchette, 2012; Sullivan et al., 2014).

The use of network analysis is increasingly common within team sport performance analysis as a means to explore on-field player interactions based on passes performed between teammates, also known as possession chains. Network analysis has been used in this way to examine collective patterns of play in soccer and basketball (Duch et al., 2010; Clemente, Lourenço Martins et al., 2015; Fewell et al., 2012; Gonçalves et al., 2017; Grund, 2012). In these studies, this technique has been used to evaluate group stability, identify influential individuals, and quantify the predictability of interactions between individuals within a team. Common network metrics that are relevant to team possession chain interactions include degree centrality,

network density, team entropy, and cluster coefficients. Teams will display high degree centrality when there is a small number of players who receive and pass the ball most frequently, and are influential “nodes” within a team's offensive structure (Clemente, Couceiro et al., 2015; Clemente, Lourenço Martins et al., 2015; Malta & Travassos, 2014). Teams that complete a higher number of successful passes per possession chain will display higher network density (Gonçalves et al., 2017). Teams with high network entropy employ irregular ball movement patterns that can create confusion for opponents and such teams are thought to be more difficult to defend due to this unpredictability (Fewell et al., 2012). Clustered networks represent greater interaction between players and can reflect localised cooperation (Clemente, Lourenço Martins et al., 2015; Peña & Touchette, 2012).

There is growing research interest in evaluating the ways in which network dynamics might reflect collective actions that are associated with team success. Studies in soccer and basketball have observed that highly centralised team networks were associated with lower entropy, lower cluster coefficient, and lower density, which in turn were associated with less successful team performances (Clemente, Couceiro et al., 2015; Clemente, Lourenço Martins et al., 2015; Fewell et al., 2012). These results indicate that, if teams build their possession chains around a small selection of key players, with weak interactions within and between clusters, and fewer disposals shared amongst those involved, their ball movement is more predictable to opposing sides and is likely to result in less successful outcomes. Conversely, teams that show greater

distribution of possession amongst team members create larger clusters of “interconnectivity” (Clemente, Lourenço Martins et al., 2015). This can create more complex, less predictable patterns of play.

Australian football has several unique characteristics that lend itself to network analysis. There are many opportunities for in-game player interactions due to: the number of players on-field at any one time (18 per team in the Australian Football League (AFL)); the dynamic and free-flowing nature of game play; and the nomadic, sometimes overlapping roles of players in different field positions (Gray & Jenkins, 2010). In Australian football, interactions between players during offensive passages are most easily observed in passes of the ball, via kicks and handballs, executed by one player and received by another. A sequence of these passes by a team is referred to as a “possession chain”, with each chain starting when that team gains possession (e.g., after an opposition score or turnover, or the ball is deemed out of bounds) and ending when the same team loses possession.

Braham and Small (2018) recently employed network analysis techniques to possession chain data from the 2014 AFL season. A range of network metrics were generated (e.g., betweenness, a measure of centrality; team entropy), which were then included in candidate models for predicting match outcomes, alongside a selection of traditional game statistics (e.g., number of disposals, shot efficiency). Using a sports betting paradigm, the candidate models were compared to each other and to a reference model that predicted match outcomes in line with bookmakers’ favourites and margins, with each model having hypothetically placed \$1 bets on each match. The reference model was the most accurate for picking winners (72.6% correctly predicted), but it also recorded a net loss of −\$7.65 across the season’s predictions. All 8 candidate models that included network centrality measures outperformed the reference model in terms of net returns, with 6 of these models making net profits ranging from \$1.94 to \$26.46. Of the 6 models that included traditional game statistics, only 2 returned net profits across the season’s predictions (model including disposals: \$0.18; model including passing rate: \$12.22). Overall, these results indicate that network metrics contain additional information about AFL team performance that is not captured by traditional game statistics, nor in the subjective assessments of favouritism made by participants in the betting market. Opportunities remain for further research conducted through a performance analysis lens, where the analytical intent is to understand how Australian football teams work together as collective units to achieve shared goals and successfully execute their tactical plans. Such insights would be valuable to coaches and performance analysts seeking evidence to guide their design, practice, and implementation of coordinated actions and tactical structures (Ribeiro et al., 2017).

The aim of this study was to investigate on-field player interactions in the Australian Football League by analysing the network characteristics of kick-in possession chains, and examining their effects on kick-in outcomes, match outcomes, and end-of-season ladder position. Kick-in chains take place in a relatively structured scenario and always start from the defensive goal square, which provides a useful in-game scenario for evaluating the utility of network analysis for understanding intentional ball movement tactics

in Australian football. It was hypothesised that less predictable patterns of play (measured as high team entropy, high network density, and low centrality) would more frequently lead to desirable chain outcomes (e.g., scoring a goal for 6 points vs. scoring a behind for 1 point vs. committing a turnover) and be exhibited by more successful teams (e.g., top 6 on the end-of-season ladder, winning teams). Similarly, it was hypothesised that greater predictability in possession chains would be associated with less successful chain, match, and season outcomes.

Methods

Sample

This study used commercially available data from the 2015 AFL season supplied by Champion Data, the official statistics provider for the AFL. An exemption from ethics review was granted by the institutional ethics committee. The original data set included all 149,449 possession events from all matches played during the 2015 AFL season (18 teams, 665 players, 206 regular season and finals series games). A data subset was created to focus the analysis on regular season matches in order to explore whole-of-league patterns (i.e., finals matches were omitted as they are only played by teams in the top 8 end-of-season ladder positions). This subset was filtered further to include only those sequences of possession events that i) started with a kick-in following a behind, and ii) ended with the offensive team losing possession through a score, stoppage, or turnover. These sequences are referred to herein as “kick-in possession chains”, where a sequence begins with the player who first receives the ball from a kick-in and excludes the initial disposal out of the goal square, unless the player starting with possession “plays on” to himself. Possession chains that involved only one player or were incomplete due to end-of-quarter time restrictions were excluded. Chains from 2 drawn matches were also excluded. The final data set used for analysis comprised a total of 1,720 kick-in possession chains from 194 games.

Measures, variables, and data preparation

To quantify network metrics from the possession interactions between players, discrete disposal and possession event data was combined to create an edge list, which consisted of two columns of data: one column listing the “source” and the other listing the “target” of each interaction between two players. In the context of this study, the “source” is the person disposing of the ball, and the “target” is the person receiving possession of the ball. The edge list was then weighted to reflect the number of times any two players interacted with each other. For example, a weighting of one would be assigned to two players that only interacted once out of all kick-in chains in one match. The weighted edge list was converted into an adjacency matrix, which was then used in the network analysis process to quantify the network metrics described in Table 1 (all unit-less indices).

Team outcomes were expressed in three ways: kick-in chain outcomes, match outcomes, and season outcomes. The outcomes of each kick-in possession chain were coded into 10 discrete categories based on the end-location (see Figure 1)

Table 1. Summary of network metrics used in this study to evaluate player interactions within possession networks.

Network metric	Use in this study	Typical values
Degree centrality	Measures the number of ties one node has to other nodes. In this study, this represents the frequency of player involvement in kick-in chains (Clemente, Couceiro et al., 2015).	Range from 0 to 1. Higher values indicating a more centralised network (Clemente, Couceiro et al., 2015).
Density	The number of observed interactions between players, divided by the total possible number of interactions between all players involved in a given possession chain (Clemente, Couceiro et al., 2015).	Range from 0 to 1. Values closer to 1 indicate interactions between all possible players, whereas lower values (~0.5) represent fewer connections (Clemente, Couceiro et al., 2015).
Cluster coefficient	Measures the degree of connection between players within a team (Fewell et al., 2012), reflecting subgroups who frequently coordinate through exchanging possession of the ball (Ribeiro et al., 2017).	Range from 0 to 1. Values closer to one indicate greater connectedness between players (Fewell et al., 2012).
Team entropy	Measures the predictability of ball movement between playing positions (Fewell et al., 2012).	Typical values vary depending on the constraints of the network. In this study, lower values indicate more predictable ball movement and higher values indicate less predictable ball movement.

and result of the chain as recorded by Champion Data (Jackson, 2016). These categories are, in order from most to least successful: goal, behind, forward 50 stoppage, forward 50 turnover, attacking midfield stoppage, attacking midfield turnover, defensive midfield stoppage, defensive midfield turnover, defensive 50 stoppage, defensive 50 turnover. Match outcomes were expressed as whether the team in possession ended up winning or losing a game. Season outcomes were represented by position on the end-of-season ladder, with teams grouped into brackets of top 6, middle 6, and bottom 6 (i.e., tertile split based on final ladder positions) (Altman & Bland, 1994).

Statistical analysis

Initial tests confirmed that the data met assumptions of normality (Fisher's coefficients for skewness and kurtosis within the range -1.96 to +1.96) (Pett, 2015) and homoscedasticity (Levene's test returns a non-significant F value where $p < 0.05$) (Levene, 1960). Descriptive statistics (e.g., means and standard deviations) for each of these metrics were reported for each kick-in outcome, for each ladder bracket (i.e., top 6, middle 6, bottom 6), and for each match outcome (i.e., win, loss). A three-way analysis of variance (ANOVA) was conducted for each of the four metrics, to determine whether these network characteristics differed due to kick-in outcome, match outcome, and ladder bracket. Main effects and two-factor interaction effects were assessed via the ANOVA procedure. Where

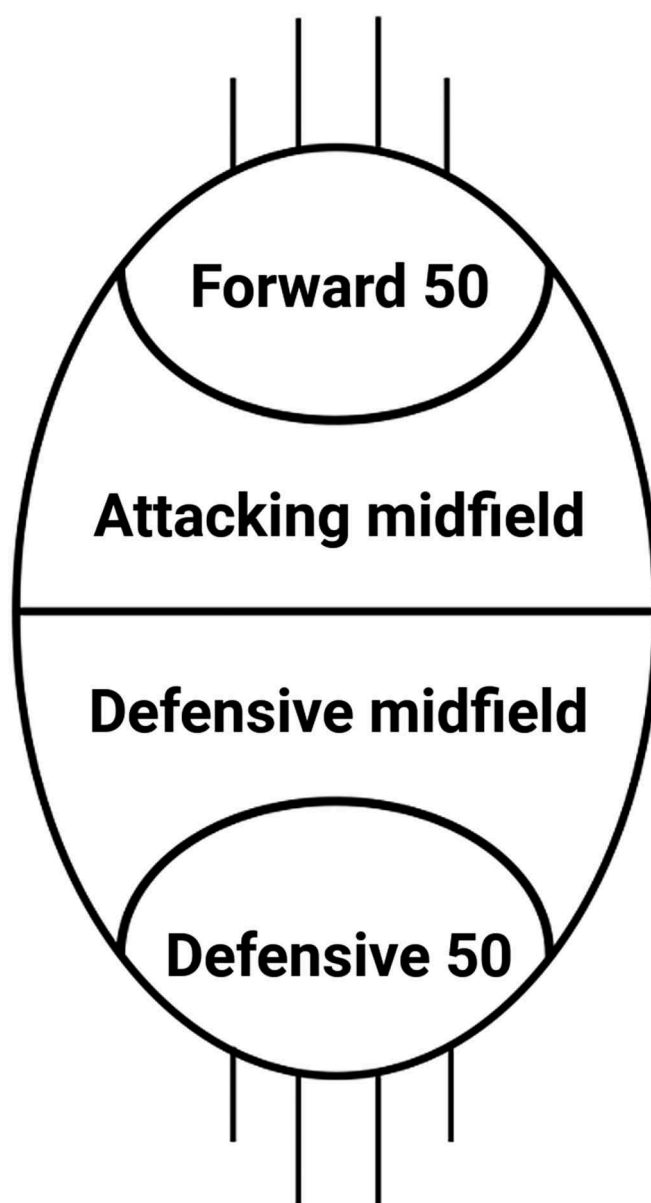


Figure 1. Major field zones in Australian football.

significant main or interaction effects were identified for factors with more than 2 levels, Tukey's Honestly Significant Difference post hoc tests were conducted to locate pairwise differences. The omega squared (ω^2) statistic was used to quantify effect sizes within the factorial ANOVA (Lakens, 2013). Effect sizes were interpreted in accordance with the thresholds proposed by Cohen (1988): $\omega^2 \leq 0.01$, small; $0.01 > \omega^2 \leq 0.06$, medium; $\omega^2 > 0.06$, large. The three-way interaction term was omitted from the ANOVA procedure, on the basis that it was unclear whether such an interaction effect was interpretable in a practically meaningful way. All data preparation and statistical analysis procedures were conducted in R (R Core Team, 2018) using the following packages: here (Müller, 2017), readr (Wickham et al., 2017), dplyr (Wickham et al., 2018), tidyr (Wickham & Henry, 2018), stringr (Wickham, 2018), purrr (Henry & Wickham, 2018), stringi (Gagolewski, 2018), reshape2 (Wickham, 2007), igraph (Csardi & Nepusz, 2006), network (Butts, 2008), entropy (Hausser & Strimmer, 2014), fitdistrplus

(Delignette-Muller & Dutang, 2015), pastecs (Grosjean & Ibanez, 2018), car (Fox & Weisberg, 2011), sjstats (Lüdtke, 2018), rmarkdown (Allaire et al., 2018), flexdashboard (Iannone et al., 2018), knitr (Xie, 2015), formattable (Ren & Russell, 2016).

Results

Descriptive statistics are presented in Table 2. Figure 2 illustrates how each of the four network metrics varied with different kick-in outcomes. Cluster coefficients were comparable across kick-in outcomes, suggesting that small, similarly sized clusters form within kick-in possession chains irrespective of

how and where chains end. Low centralisation was recorded across all 10 kick-in outcomes, with little variability observed between outcomes. This suggests that, following a kick-in, possession of the ball was generally distributed amongst several players, and again this was irrespective of how and where chains ended. The highest network density values were observed for chains that ended in the least desirable kick-in outcomes; high network density values indicate less direct ball movement. Conversely, the most direct ball movement patterns (i.e., low network density values) were observed for the most desirable outcomes. More successful kick-in chains exhibited less predictable ball movement patterns as indicated by

Table 2. Descriptive statistics for the four selected network metrics, categorised by kick-in outcome, ladder bracket, and match outcome (mean $\pm$ standard deviation).

	Cluster coefficient	Degree centrality	Network density	Entropy
Kick-in outcome				
Goal	0.070 $\pm$ 0.134	0.107 $\pm$ 0.076	0.165 $\pm$ 0.055	2.267 $\pm$ 0.366
Behind	0.076 $\pm$ 0.147	0.118 $\pm$ 0.091	0.177 $\pm$ 0.077	2.219 $\pm$ 0.369
Forward stoppage	0.068 $\pm$ 0.176	0.135 $\pm$ 0.102	0.205 $\pm$ 0.073	2.038 $\pm$ 0.318
Forward turnover	0.072 $\pm$ 0.165	0.129 $\pm$ 0.089	0.193 $\pm$ 0.086	2.200 $\pm$ 0.452
Attacking midfield stoppage	0.054 $\pm$ 0.140	0.126 $\pm$ 0.100	0.260 $\pm$ 0.114	1.912 $\pm$ 0.366
Attacking midfield turnover	0.080 $\pm$ 0.171	0.128 $\pm$ 0.087	0.225 $\pm$ 0.105	2.098 $\pm$ 0.452
Defensive midfield stoppage	0.087 $\pm$ 0.206	0.100 $\pm$ 0.108	0.336 $\pm$ 0.137	1.773 $\pm$ 0.401
Defensive midfield turnover	0.072 $\pm$ 0.163	0.129 $\pm$ 0.110	0.262 $\pm$ 0.125	2.044 $\pm$ 0.471
Defensive stoppage	0.000 $\pm$ 0.000	0.090 $\pm$ 0.119	0.412 $\pm$ 0.105	1.493 $\pm$ 0.170
Defensive turnover	0.088 $\pm$ 0.188	0.099 $\pm$ 0.126	0.378 $\pm$ 0.123	1.622 $\pm$ 0.311
Ladder bracket				
Top 6	0.083 $\pm$ 0.184	0.116 $\pm$ 0.096	0.242 $\pm$ 0.123	2.037 $\pm$ 0.455
Middle 6	0.064 $\pm$ 0.150	0.126 $\pm$ 0.105	0.246 $\pm$ 0.123	2.028 $\pm$ 0.448
Bottom 6	0.071 $\pm$ 0.154	0.118 $\pm$ 0.097	0.245 $\pm$ 0.124	2.035 $\pm$ 0.452
Match outcome				
Win	0.072 $\pm$ 0.162	0.122 $\pm$ 0.104	0.248 $\pm$ 0.124	2.025 $\pm$ 0.448
Loss	0.073 $\pm$ 0.165	0.118 $\pm$ 0.095	0.242 $\pm$ 0.122	2.041 $\pm$ 0.455

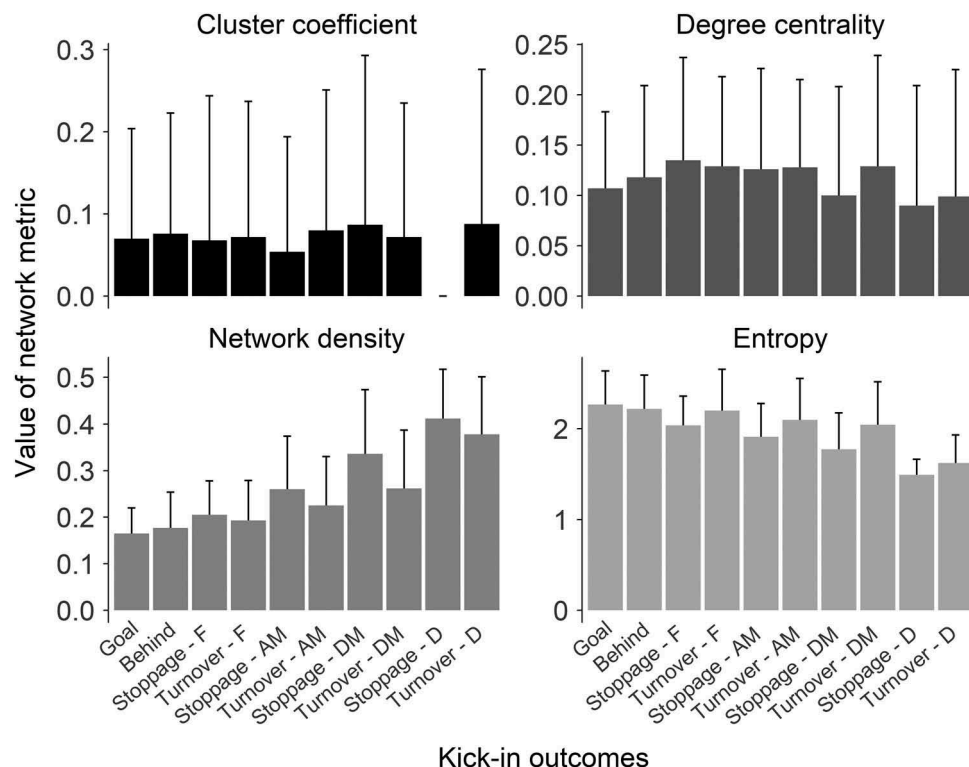


Figure 2. Network metrics for kick-in outcomes (mean $\pm$ standard deviation). Note: F = Forward 50, AM = Attacking midfield, DM = Defensive midfield, D = Defensive 50.

high entropy values. In contrast, the least successful possession chains were characterised by more predictable ball movement, evidenced by low entropy values. For each of the four network metrics, there were no notable differences between the ladder brackets (top 6 vs. middle 6 vs. bottom 6) nor the two match outcomes (win vs. loss).

There were no significant main or interaction effects identified by the three-way ANOVA for cluster coefficient. For degree centrality, a significant main effect for kick-in outcome was observed ($F(9, 1678) = 3.01, p < 0.01$). However, the magnitude of this omnibus effect was small ($\omega^2 = 0.01$) and only one significant pairwise difference was identified from the Tukey post hoc test. Thus, neither of these network metrics showed notable variation by kick-in outcome, match outcome, or season outcome.

For network density, a significant main effect was observed for kick-in outcome ($F(9, 1678) = 66.6, p < 0.00$). The omnibus effect for kick-in outcome was large in magnitude ($\omega^2 = 0.26$), and 35 out of 45 pairwise comparisons were significantly different based on post hoc testing (Figure 3). The overarching pattern shown in Table 2 is that kick-in chains that ended in more desirable outcomes exhibited significantly lower network density (i.e., more direct ball movement), compared to chains that ended in less desirable outcomes. For instance, chains ending in a goal exhibited significantly lower density values compared to chains ending in the attacking midfield, defensive midfield, and defensive 50 field zones. A significant main effect was also observed for match outcome ($F(1, 1678) = 6.46, p = 0.01$) but the size of this effect was near-zero in magnitude ($\omega^2 < 0.01$).

For entropy, a significant main effect was identified for kick-in outcome ($F(9, 1678) = 39.6, p < 0.00$). This main effect was large in magnitude ($\omega^2 = 0.17$), with post hoc testing identifying that 34 out of 45 pairwise comparisons were significantly different (Figure 4). Considered alongside the results presented in Table 2, the analysis showed that kick-in chains ending in more desirable outcomes were characterised by higher entropy values (i.e., lower predictability), compared to chains that

ended in less desirable outcomes. For example, entropy was significantly higher for chains ending in a goal, compared to chains ending in a forward 50 stoppage, or in the attacking midfield, defensive midfield, and defensive 50 field zones.

Discussion

Contrary to what was hypothesised, network characteristics did not differ between teams in different ladder brackets nor between teams with different match outcomes. However, the results indicate that kick-in possession chains ending in more desirable outcomes exhibit low network density and high entropy (i.e., more direct and less predictable patterns of ball movement following kick-ins). This study shows the promise of applying network analysis techniques for obtaining quantitative insights into how AFL players interact in order to distribute possession of the ball within their teams.

No notable differences between kick-in outcomes, match outcomes, nor team outcomes were observed for degree centrality and cluster coefficient. These observations differ somewhat from previous studies that have observed associations between other centrality measures and measures of team success. For instance, Braham and Small (2018) reported that two measures of centrality—betweenness and mean out-degree—provided additional information beyond conventional game statistics (e.g., number of disposals, number of clearances) when modelling AFL team score and score margin between opposing teams. These researchers also reported a small, significant difference between winning and losing teams, with 56.3% of winning teams recording higher cluster coefficient values than their opposition. Studies in other sports have shown that passing networks with higher cluster coefficients (i.e., more interconnectedness) are associated with successful outcomes in basketball and soccer (Clemente, Couceiro et al., 2015; Fewell et al., 2012). The differences between the Braham and Small (2018) study and our results may be due to the targeted focus on kick-in possession chains in our research. As a consequence, generalising results from the present study to

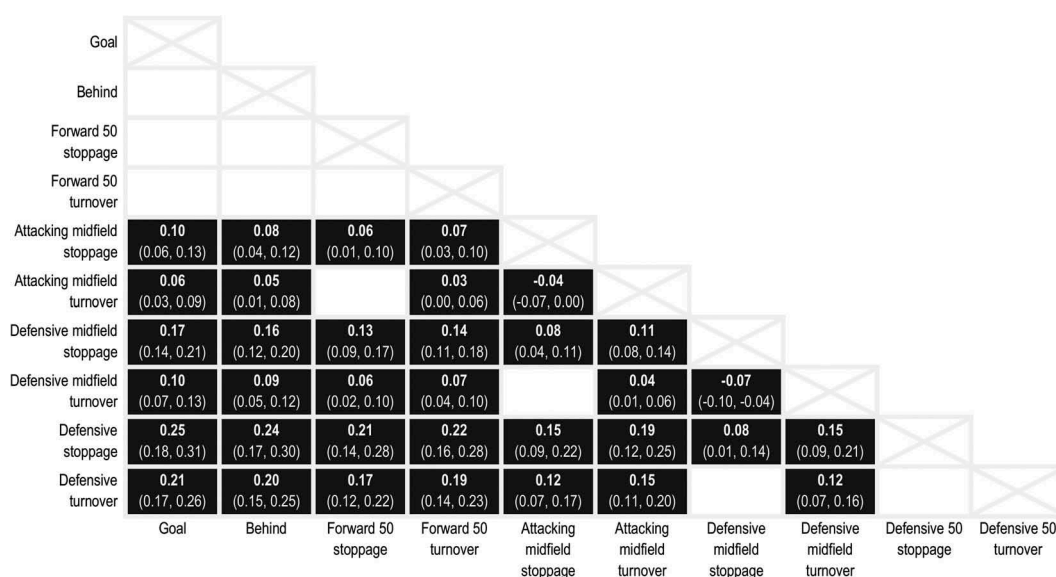


Figure 3. Mean differences (with 95% confidence limits: lower, upper) in network density between pairs of kick-in outcomes. Notes: Mean difference = y variable minus x variable. Grey cells denote significant pairwise differences, blank cells denote no significant pairwise difference.

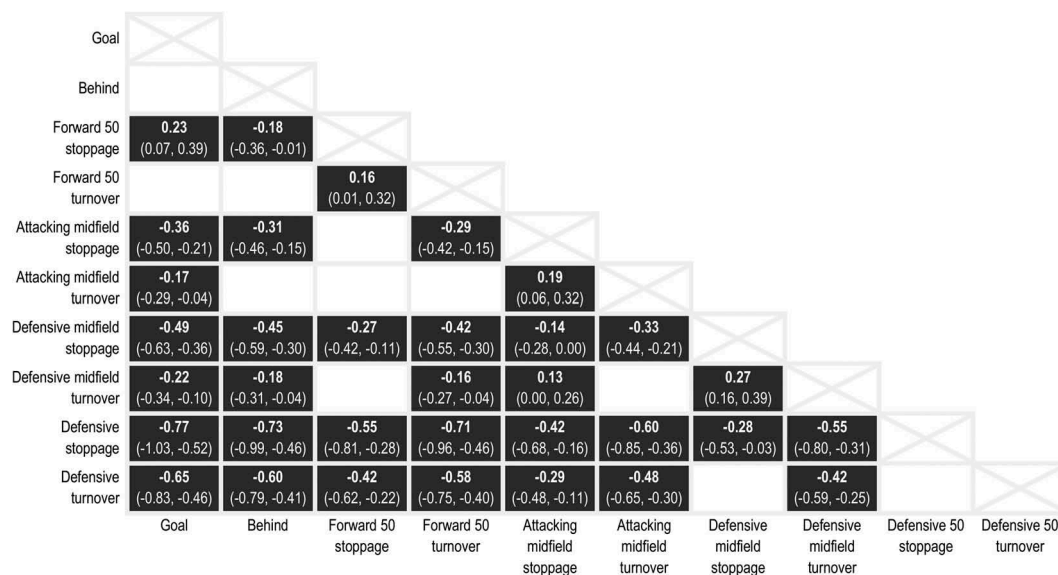


Figure 4. Mean differences (with 95% confidence limits: lower, upper) for significant differences in entropy between pairs of kick-in outcomes. Notes: Mean difference = y variable minus x variable. Grey cells denote significant pairwise differences, blank cells denote no significant pairwise difference.

other passages of play may not be appropriate. Moreover, characteristics of successful passing networks are likely to be influenced by the rules of each sport, so insights about the characteristics of successful possession networks in one sport may not always be transferable to other sports. We encourage further research that explores whether different passages of play in Australian football exhibit unique network signatures that are related to team performance outcomes.

Network density was significantly lower for kick-in chains ending in more desirable outcomes compared to those that ended in less desirable outcomes, in contrast to our hypothesis. The results of the present study agree with Clemente et al. (2014), who observed that national soccer teams employed more direct passing patterns with low network density when counter-attacking after an opposition goal. Conversely, previous literature has observed that more successful teams generally exhibit high network density in their passing interactions (Clemente, Couceiro et al., 2015; Clemente, Lourenço Martins et al., 2015; Grund, 2012). These results may suggest that teams exhibit low network density when moving the ball from defence potentially due to the use of more attacking and direct styles of play, but that when possession networks in other field zones are considered, it is advantageous to execute a passing strategy that involves a wider variety of players who are closely interconnected. Alternatively, chains that ended in defensive zones may display higher network density due to the pressure imposed by the opposition (i.e., the team that is not in possession). Contemporary styles of Australian football exhibit a “re-possession style of football” (Woods et al., 2017), where teams implement full-ground defensive structures and apply what is colloquially known as “forward pressure” as they attempt to regain possession of the ball. In response, the attacking team may pass the ball between many players—thereby increasing the density of their passing network—in an attempt to create and take advantage of weaknesses in the opposition’s defensive structure.

The discrepancies between the results of the present study and previous research may be explained by differences in the

rules between various football codes. In soccer, the offside rule does not allow offensive players to position themselves beyond the last line of field defenders, and thus limits forward movement of the ball (Ford, 2008). In addition, fewer players take the field in soccer (11 per team, compared to 18 per team in AFL), thus more passes are generally executed amongst the same group of players to progress the ball forward towards the goal, requiring more interconnectedness within teams. Rather than having to pass the ball back and forth between a small number of players, the rules of AFL afford teams with the opportunity to involve a greater number of individuals when moving the ball and progressing it towards the goals. This allows AFL players the option to advance the ball down the field using more direct passing sequences in order to transfer ball possession to players already positioned close to the goals. It is also important to note that the density metric is dependent on the number of players involved in possession as well as the total number of interactions (Clemente, Lourenço Martins et al., 2015). Thus, team sports consisting of more players may complete a large number of passes, but if all possible interactions are not realised, network density is decreased. In the context of AFL, it might not be necessary nor optimal to strive to achieve high interconnectedness between all players involved in kick-in possession chains, since forward movement of the ball is not limited by the rules of the sport.

The types of passes used in Australian football may also explain the discrepancies between our results and previous research conducted in soccer. In Australian football, the ball can be disposed of via kicks or handballs, which provides options for controlling the pace of the game. For example, a player that marks the ball (catching the ball in the air after a team-mate’s kick and before the ball bounces) can choose to slow the team’s ball movement down. The mark typically affords a few seconds of time and several metres of space away from defenders, who are not allowed to encroach beyond where the ball was marked to close the distance between themselves and the ball carrier. Thus, pace of ball movement

is one facet of offensive playing style which can be manipulated strategically and situationally in Australian football. In turn, the offensive style employed by a team is likely to influence network density, since the team's intended and implemented tactics will affect the number and speed of disposals executed (Clemente, Couceiro et al., 2015). Researchers investigating possession networks in AFL may seek to explore this further by incorporating the type of disposal (i.e., kick or handball) and the mode of possession (i.e., mark or "play on") in their analysis.

Entropy was significantly higher for kick-in chains ending in more successful outcomes, supporting our hypothesis that less predictable ball movement patterns are associated with more successful kick-in outcomes in the AFL. Our results are consistent with the work of Braham and Small (2018), who demonstrated that 64.6% of winning AFL teams also exhibit high entropy values, a proportion that was significantly different to the null distribution. In a separate analysis, the same researchers observed that the inclusion of team entropy (alongside centrality metrics, as described earlier) to conventional AFL game statistics improved model fit. Specifically, when compared to the regression including only conventional AFL metrics, adding network metrics resulted in r^2 increasing from 0.588 to 0.622 (difference = +0.034), while the Bayesian information criterion decreased from 3597.9 to 3580.4 (difference = -17.5) (Braham & Small, 2018). Fewell et al. (2012) also reported an association between high entropy and tournament advancement in basketball networks. In team sports, the ability to move the ball in irregular patterns is thought to provide a competitive advantage by allowing teams to evade and disorganise the opposition's defensive tactics (Fewell et al., 2012; Hewitt et al., 2016). Our results contribute to an emerging evidence base demonstrating that the degree to which passing networks are predictable is related to how successful these passing sequences are. These findings also highlight the value of network analysis as a means to objectively assess the degree to which teams employ predictable patterns of play (D'Amour et al., 2015).

Limitations and future directions

The present study focused solely on kick-in possession chains. Kick-in chains only account for a small proportion of possession chains (8% of all possession events in the 2015 home and away season), thus it is important that the presented findings are not generalised to other phases of attacking play in elite Australian football. In addition, the most successful AFL teams may not be the most accomplished in kick-in ball movement. Future research is warranted to examine more complex scenarios in Australian football, such as centre clearances, which lead to a greater proportion of scores (Easton-Corke & Luff, 2016; Stewart et al., 2007). It may be worth controlling for the number of passes in a possession chain in order to explore the network dynamics that emerge from different possession styles of AFL teams (e.g., possession chains that cover more goal-to-goal distance with fewer passes might represent direct styles of play, while chains comprising more passes might represent a high-possession and indirect style of play). Grouping the sample into brackets may have also limited the sensitivity of the present analysis for detecting the differences between highly successful and highly unsuccessful teams. Other factors that could affect the possession networks of AFL teams

include the strength and style of the opposition's defence, with defensive actions in Australian football more difficult to discern and accurately capture when compared to possession and disposal events. Continued advancements in positional tracking and automated video processing technology may support future research that investigates how defensive team positioning and actions affect offensive team tactics.

Practical implications

Our results demonstrate that lower density and higher entropy are associated with successful kick-ins. In practical terms, this indicates that successful ball movement patterns out of a kick-in are typified by direct passes and the involvement of a wide variety of players in the passing networks (thereby decreasing the predictability of player interactions). Given the dynamic nature of offensive and defensive game play in AFL, these proposed implications may best serve coaches and players when considered as principles that can be practiced and adapted to suit varied kick-in scenarios, rather than as a rigid formula for optimising ball movement. These findings and the analytical approach may also be useful from a defensive perspective. For example, coaches and performance analysts could utilise the same methods to analyse other teams, in order to identify ways to regain possession within the forward 50 during an opposition kick-in (Appleby and Dawson, 2002).

Conclusion

Our findings indicate that successful and unsuccessful AFL teams do not significantly differ with respect to their ball movement networks following kick-ins. This may reflect the high variability in offensive playing styles and kick-in tactics employed by teams of varying success levels. The analysis did reveal that kick-in chains leading to more desirable outcomes were characterised by lower density and higher entropy; that is, more direct and less predictable patterns of possession. This provides quantitative evidence for Australian football coaches and performance analysts that may enhance their understanding of effective kick-in tactics employed by AFL teams. This research adds to the growing evidence base that demonstrates the utility of network analysis for investigating patterns of possession play, and provides a platform for further applications of this analytical approach to other aspects of game-play in Australian football (e.g., stoppages, transition offence, transition defence).

Disclosure Statement

No potential conflict of interest was reported by the authors.

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